

DAYANANDA SAGAR COLLEGE OF ENGINEERING

**Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning**



INTERNSHIP REPORT

ON

“Backend Development”

Submitted in partial fulfillment for the award of degree(20AI8ICINT)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

ANKUR SINGH - 1DS20AI008

Conducted at

DailyPe

DailyPe

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “ **Backend Development** ” carried out by **Mr Ankur Singh** bearing USN **1DS20AI008**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2023-2024. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (**20AI8ICINT**)

Signature of Reporting Head
Saurabh Gupta
Co-Founder
DailyPe

Signature of Internship Coordinator
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P
Malagi Dept. of AI &
ML DSCE

External Viva:

Name of the Examiner

1) _____

2) _____

Signature with Date

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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DECLARATION

I, **Ankur Singh**, final year student of Information Science and Engineering, Dayananda Sagar College Of Engineering, Bangalore - 560078, declare that the Internship has been successfully completed, in **DailyPe**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor Degree in Artificial Intelligence and Machine Learning, during the academic year 2023- 2024.

Date:

NAME: ANKUR SINGH

Place: Bangalore

USN: 1DS20AI008

OFFER LETTER

NEOFIRST TECHNOLOGIES INDIA PRIVATE LIMITED

CIN: U72200KA2023FTC170246

Reg. Address - No. 806 B2, Alpine ECO Apts, Marathalli, Doddanekkundi, Bangalore - 560037

E-mail - saurabhgupta2104@gmail.com. Mobile No.: 7737557609

INTERNSHIP CONTRACT AGREEMENT

THIS INTERNSHIP CONTRACT AGREEMENT (HEREINAFTER REFERRED TO AS THE "AGREEMENT") IS ENTERED INTO ON 14TH August, 2023 (THE "EFFECTIVE DATE"),

BY AND BETWEEN

M/s. Neofirst Technologies India Private Limited, (CIN: U72200KA2023FTC170246) having its registered office at No. 806, B2, Alpine Eco Apts, Doddanekkundi, Marathalli, Bangalore, Kamataka - 560037, (hereinafter referred to as the "Company") on the **FIRST PART**;

AND

Mr. Ankur Singh, (PAN: NSFPS0195I) S/o Mr. Mithilesh Kumar Singh, aged around 21 years residing at House No 6-88/63, Renuka Residency, Beside Essar Petrol Pump, Annaram, Sangareddy, Telengana- 502313, (hereinafter referred to as the "Intern") on the **SECOND PART**

Both the Company and Intern hereinafter collectively referred to as the "Parties"

Whereas the Company is involved in the business of IT - Software solutions.

Whereas the Intern who was in need of the internship has approached the Company and the Company has agreed for the same.

Now the parties would like to reduce terms and condition into writing hence, this agreement.

NOW THIS AGREEMENT WITNESSTH AS FOLLOWS

1. INTERNSHIP POSITION

The Intern has been assigned to the position of an Intern in the Backend Department.

COMPLETION CERTIFICATE

NEOFIRST TECHNOLOGIES INDIA PRIVATE LIMITED

CIN: U72200KA2023FTC170246

Reg. Address - No. 806 B2, Alpine ECO Apts, Marathalli, Doddanekkundi, Bangalore - 560037

E-mail - saurabhgupta2104@gmail.com, Mobile No.: [7737557609](tel:7737557609)

Date: 10-04-2024

TO WHOM IT MAY CONCERN

This is to confirm that Mr. Ankur Singh, a student of **B.E in Artificial Intelligence & Machine Learning (Final Year)** Dayananda Sagar College of Engineering, Bangalore, has completed 7 months of an internship program from **Aug 2023 to Mar 2024** on **Backend Services**.

He was highly motivated & hardworking. He worked sincerely at his tasks & did an excellent job. We are delighted with the performance and the speed of learning, implementing things was amazing.

We wish him great success in his future endeavors.



Authorized Signatures

Thanks & Regards,

Saurabh Gupta

Co-Founder

Neofirst Technologies India Pvt. Ltd.

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal Dr. B G Prasad, for providing us adequate facilities to undertake this Internship.

We would like to thank Dr. Vindhya P Malagi, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank Mr Saurabh Gupta, DailyPe, for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, Prof. Kavya D N, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

ANKUR SINGH
1DS20AI008

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CHAPTER 1

INTRODUCTION

DailyPe is a dynamic marketplace catering to the financial needs of businesses, particularly those requiring small-term loans with streamlined procedures. Unlike traditional lending avenues burdened with extensive paperwork, DailyPe offers a convenient solution, minimizing bureaucratic hurdles. One distinguishing feature is its repayment structure, which aligns with the rhythm of daily business operations. Through this model, merchants repay loans in manageable installments, easing cash flow concerns and promoting financial stability. Additionally, the platform integrates seamlessly with digital wallets, facilitating swift transactions and ensuring efficient loan repayment.

At the core of DailyPe's functionality is its commitment to empowering businesses through accessible financial solutions. By requiring installment payments both at month-end and during daily transactions, the platform fosters a symbiotic relationship between lending and commerce. This innovative approach not only provides businesses with the necessary capital to thrive but also reinforces financial discipline. Moreover, the daily repayment mechanism reflects an understanding of the ebb and flow of business activities, aligning loan obligations with revenue generation. As a result, DailyPe emerges as a vital ally for merchants, offering not just financial assistance but also a strategic partnership geared towards sustained growth.

ABOUT CO-FOUNDER

Saurabh Gupta

He has 2 years of experience building products in the lending tech domain at FinBox. He created products such as Device Connect and Bank Statement Analyzer to evaluate New to Credit customers. During his college years, he opened a glass bottle recycling factory.

Shashwat Shrikhande

He enjoys working on problems related to the lending tech industry and wants to help reduce the credit gap in the Indian economy. He graduated from IIT Kharagpur in 2021 and then worked at ICICI Bank & Lenskart.com before starting up DailyPe. He is an ardent Chelsea Football Club fan.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

DailyPe stands as a dynamic marketplace that caters specifically to the financial needs of businesses, particularly those seeking small-term loans with a focus on efficiency and simplicity. In contrast to the cumbersome processes often associated with traditional lending institutions, DailyPe offers a streamlined solution that minimizes bureaucratic hurdles. This emphasis on simplicity extends throughout the platform's operations, from the initial application process to loan repayment.

One of the standout features of DailyPe is its innovative repayment structure, which is intricately designed to synchronize with the daily rhythm of business operations. Through this model, merchants are relieved of the burden of making large lump-sum payments, instead opting for manageable installments that align with their cash flow. By easing cash flow concerns and promoting financial stability, DailyPe empowers businesses to focus on growth and innovation rather than being bogged down by financial constraints. Furthermore, the platform's seamless integration with digital wallets enhances convenience for users, facilitating swift transactions and ensuring efficient loan repayment.

At the heart of DailyPe's functionality lies a profound commitment to empowering businesses through accessible financial solutions. Beyond simply providing capital, DailyPe fosters a symbiotic relationship between lending and commerce by requiring installment payments both at month-end and during daily transactions. This innovative approach not only addresses immediate financial needs but also instills a sense of financial discipline among borrowers. By aligning loan obligations with revenue generation, DailyPe demonstrates a keen understanding of the ebb and flow of business activities, positioning itself as a strategic ally for merchants seeking sustained growth and success.

CHAPTER 3

SCOPE OF WORK

Loan Processing and Management

DailyPe's team is responsible for efficiently processing loan applications, assessing eligibility criteria, and managing the entire lifecycle of loans, from origination to repayment. This involves conducting thorough due diligence to evaluate creditworthiness and mitigate risks associated with lending.

Technology and Platform Development

The development and maintenance of DailyPe's technological infrastructure are critical for ensuring a seamless user experience. This includes the continuous enhancement of the platform's functionality, security measures, and integration with digital payment systems.

Customer Support and Relationship Management

DailyPe places a strong emphasis on providing exceptional customer support to its users. This entails addressing inquiries, resolving issues, and maintaining positive relationships with borrowers, ensuring their satisfaction and loyalty to the platform.

Risk Management and Compliance

Given the nature of financial services, DailyPe must adhere to strict regulatory requirements and implement robust risk management practices. This involves monitoring and mitigating various types of risks, such as credit risk, operational risk, and compliance risk, to safeguard the interests of both the company and its customers.

Business Development and Partnerships

DailyPe's growth and expansion rely on forging strategic partnerships and alliances with other businesses, financial institutions, and industry stakeholders. This involves identifying opportunities for collaboration, negotiating agreements, and implementing initiatives to broaden DailyPe's reach and impact in the market.

CHAPTER 4

LEARNING OBJECTIVES AS GIVEN BY COMPANY

DailyPe has recognized my potential and designed my internship experience to equip me with a valuable skill set for full-stack development in the fintech environment. Here's a detailed breakdown of their learning objectives:

Gaining Practical Experience in AWS

During my time at DailyPe, I've had the opportunity to delve into various Amazon Web Services (AWS) offerings, each playing a crucial role in our operations. DynamoDB has been instrumental in managing our databases efficiently, ensuring seamless data storage and retrieval for our applications. S3 has served as a reliable solution for storing and serving static assets, enabling fast and scalable content delivery to our users. Leveraging SES (Simple Email Service), we've streamlined our email communications, sending out notifications and updates to our customers reliably and securely.

Gaining Practicle Experience in SQL

During my internship at DailyPe, I've had the invaluable opportunity to gain proficiency in SQL (Structured Query Language), a fundamental tool for managing and querying databases. Through hands-on experience and training sessions provided by DailyPe, I've learned how to write and execute SQL queries to retrieve, manipulate, and analyze data stored in our databases effectively.

Honed Communication skill

During my time at DailyPe, I've honed my communication skills through various experiences and interactions within the organization. Whether it's collaborating with team members on projects, engaging with stakeholders, or conveying complex ideas in a clear and concise manner, DailyPe has provided me with ample opportunities to strengthen my communication abilities.

Mastered Problem Solving Ability

During my journey at DailyPe, I've cultivated a robust problem-solving ability that has been instrumental in navigating various challenges and driving innovation within the organization. Through tackling complex issues and projects, I've learned to approach problems systematically, breaking them down into manageable components and devising creative solutions.

CHAPTER 5

CHALLENGES AND SOLUTIONS

Throughout my internship at DailyPe, I encountered challenges that are common experiences for developers, especially when working on a complex tasks.

1. Balancing Frontend and Backend Development

The task I was worked on likely to involved collaboration between frontend developers (focusing on the user interface) and backend developers (handling data and server-side logic). Balancing these two aspects effectively can be a hurdle, especially for an intern:

Solution

1. Regular Communication: To overcome these challenges, I likely adopted a strategy of regular communication. This might have involved daily stand-up meetings, project management tools, or dedicated communication channels to keep both teams informed and on the same page.

2. Code Reviews: Implementing code reviews can be a valuable practice for fostering collaboration and knowledge sharing. By reviewing each other's code, both frontend and backend developers gain a deeper understanding of the overall system and can identify potential issues early on.

3. Knowledge Sharing Sessions: Organizing knowledge-sharing sessions where frontend and backend developers could present their expertise to each other could have been another solution. This promotes cross-team learning and fosters a collaborative environment.

2. Learning New Technologies:

Learning new techniques can be both exciting and challenging, presenting opportunities for growth and innovation. However, it's not uncommon to encounter obstacles along the way. One common challenge is overcoming the initial learning curve associated with unfamiliar concepts or technologies.

Solution

To address this, it's helpful to break down complex topics into smaller, more digestible chunks and to seek out resources such as tutorials, online courses, or mentorship opportunities to gain a solid understanding.

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. Within my internship at DailyPe, I spearheaded the design and development of a robust Monitoring System leveraging SQL, aimed at extracting invaluable insights into customer behavior and emerging trends. By meticulously crafting SQL queries and database structures, I ensured the system's efficacy in analyzing vast datasets, thereby empowering the organization to make data-driven decisions and optimize its strategies for enhanced customer engagement and satisfaction.
2. One of my key contributions involved the implementation of an automated system utilizing AWS Lambda, a serverless computing service. This innovative solution streamlined the monitoring of data updates within our database and facilitated the seamless execution of other Lambda functions, significantly enhancing the efficiency and reliability of our data management processes. By harnessing the power of AWS Lambda, we were able to automate routine tasks, minimize manual intervention, and improve overall system scalability and resilience.
3. I took the initiative to develop a robust API tailored to efficiently retrieve bureau data of users from external sources. This API served as a pivotal bridge between our internal systems and the bureau's data repository, enabling seamless and secure data retrieval processes. Through meticulous design and implementation, I ensured that the API delivered high performance, reliability, and data integrity, thereby facilitating informed decision-making and enhancing the organization's ability to assess creditworthiness and mitigate risks effectively.
4. Significant contribution I made at DailyPe was the development of an AutoPay API designed to automate the deduction of autopay bounce charges from users' accounts. By seamlessly integrating with our payment system, this API facilitated the automatic deduction of charges in the event of failed autopay transactions, thereby streamlining the billing process and enhancing user experience. Through meticulous design and rigorous testing, I ensured that the AutoPay API operated efficiently, accurately, and securely, ultimately contributing to improved financial management and customer satisfaction within the organization.

CHAPTER 7

REFLECTION AND ANALYSIS

REFLECTION

1. **Diverse Skill Set:** The individual demonstrates proficiency in various domains, including SQL, AWS Lambda, API development, and financial systems integration. This showcases their versatility and adaptability in tackling different aspects of technology and business operations.

2. **Initiative and Leadership:** Each project mentioned involves significant initiative and leadership, indicating the individual's ability to identify opportunities for improvement and take ownership of implementing solutions. Their proactive approach is evident in developing comprehensive systems to address specific needs within the organization.

3. **Problem-Solving and Innovation:** The projects reflect a strong focus on problem-solving and innovation. For example, the development of the Monitoring System and the AutoPay API suggests a strategic approach to leveraging technology to streamline processes, enhance efficiency, and improve user experience.

ANALYSIS

Impact on Business: The projects described have tangible implications for DailyPe's operations. The Monitoring System provides valuable insights into customer behavior, which can inform decision-making and improve business strategies. Similarly, the AutoPay API automates a critical aspect of the payment process, reducing manual effort and ensuring timely and accurate billing.

Integration of Technologies: The individual effectively integrates different technologies to develop holistic solutions. For instance, combining SQL with AWS Lambda for database monitoring demonstrates a synergistic approach to leveraging cloud computing and database management technologies for enhanced performance and reliability.

Focus on Efficiency and Scalability: The emphasis on automation and API development reflects a commitment to improving efficiency and scalability within DailyPe's operations.

CHAPTER 8

RECOMMENDATIONS

Continued Learning and Skill Enhancement: Encourage the individual to continue expanding their skill set and staying updated on emerging technologies and industry trends. This could involve pursuing certifications or advanced training programs in areas such as cloud computing, database management, and software development.

Cross-Functional Collaboration: Foster opportunities for the individual to collaborate with colleagues from different departments or teams within DailyPe. Cross-functional projects can provide exposure to diverse perspectives and approaches, fostering innovation and enhancing problem-solving capabilities.

Leadership Development: Provide opportunities for the individual to take on leadership roles or mentorship responsibilities, allowing them to further develop their leadership skills and inspire others within the organization. This could involve leading project teams, organizing knowledge-sharing sessions, or participating in mentorship programs.

Feedback and Recognition: Offer regular feedback and recognition for the individual's contributions, acknowledging their efforts and achievements. This can boost morale, motivation, and job satisfaction, fostering a positive work environment and encouraging continued growth and performance excellence.

Focus on Impact and Strategic Alignment: Encourage the individual to align their projects and initiatives with the broader strategic goals and priorities of DailyPe. Emphasize the importance of delivering measurable impact and value to the organization, whether through improved efficiency, cost savings, or enhanced customer satisfaction.

CHAPTER 9

CONCLUSION

As I reflect on my journey at DailyPe and the significant contributions I've made to the organization, I'm struck by the depth of impact that each project has had. The development of the comprehensive monitoring system using SQL not only demonstrates my technical proficiency but also underscores my ability to derive valuable insights from data to inform strategic decision-making. This project has empowered DailyPe to better understand customer behavior and market trends, ultimately driving improvements in our products and services.

Moreover, the implementation of an automated system using AWS Lambda showcases my commitment to innovation and efficiency. By leveraging cutting-edge technology, I've streamlined data management processes and enhanced the reliability of our systems, positioning DailyPe for continued growth and scalability. This project represents a pivotal step forward in our journey towards embracing cloud computing and harnessing its potential to drive business success.

Additionally, the creation of APIs for efficiently fetching bureau data and automating autopay bounce charges underscores my dedication to solving real-world challenges faced by DailyPe. These solutions have not only optimized our operations but have also improved the overall user experience, fostering greater trust and satisfaction among our customers.

Looking ahead, I'm excited about the opportunities for further growth and development. I'm committed to staying abreast of emerging technologies, fostering cross-functional collaboration, and aligning my efforts with DailyPe's strategic objectives. By continuing to push the boundaries of innovation and excellence, I'm confident that I can contribute to the continued success and prosperity of DailyPe in the dynamic and competitive landscape of financial technology.

CHAPTER 10

REFERENCES

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